

AI-AUGMENTED BACKEND DEVELOPMENT

AI BlogNest API

6. Project Testing Phase

API, authentication, authorization, AI, security, and evidence testing

1. Testing Objective

The testing phase verifies that the implemented API behaves correctly, securely, and consistently under valid and invalid requests. The sample identifies Postman or Thunder Client as tools for API verification and shows testing for registration, login, blog creation, blog fetching, AI generation, and summarization.

■filecite■turn0file0■L189-L202■

2. Testing Environment

- Run the backend server using npm start.
- Use Postman or Thunder Client as the API client.
- Verify MongoDB connectivity before database-dependent tests.
- Test protected routes with valid and invalid JWT tokens.

3. Authentication Testing

- Register a new user with valid data.
- Reject invalid or incomplete registration data.
- Login with correct credentials.
- Reject incorrect credentials.
- Verify JWT issuance and validation.
- Verify protected routes reject unauthenticated requests.

4. Authorization Testing

- Verify Admin access to administrative operations.
- Verify Editor access to editorial workflows.
- Verify Author access to permitted post creation and management.
- Verify Reader access to published content and commenting.
- Verify unauthorized role escalation is blocked.

5. Blog API Testing

- Create a blog with valid content.
- Reject malformed or incomplete blog payloads.
- Fetch individual and multiple blogs.

- Update permitted blog records.
- Delete records only when authorization permits.
- Verify lifecycle status changes.

6. AI API Testing

- Send a valid thematic prompt to the blog-generation endpoint.
- Verify the response contains generated blog content.
- Test invalid or empty AI input.
- Send article content to the summarization endpoint.
- Verify a concise summary is returned.
- Verify AI routes are protected according to the project's security requirements.

7. Security Testing

- Test input validation.
- Test sanitization against malicious payloads.
- Test NoSQL parameter manipulation protection.
- Test rate limiting behavior.
- Test centralized error responses.
- Inspect structured logs for failed and successful requests.

8. Expected Testing Evidence

The sample lists User Registration, User Login, Create Blogs, Fetch Blogs, Generate AI blogs, and Summarize Blogs as API-testing demonstrations. These provide a practical evidence set for the completed backend. ■filecite■turn0file0■L193-L202■